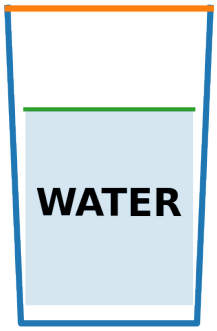
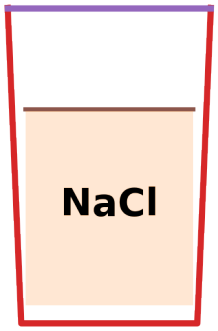
COOLING CURVES, BRINES & THE REASONING TRAP

A traditional thermal-physics investigation with an optional AI reasoning extension

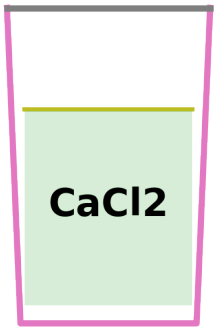
SAME STARTING TEMPERATURE • SAME CONTAINER • SAME ENVIRONMENT



250 g water



250 g water + 90 g



250 g water + 90 g

Core question

Does a lower specific heat guarantee that a solution will cool faster - or can the experimental design itself hide a more important variable?

LEVEL	TIME	MODE	DATA
Grades 8-12 / Intro college	1-2 class periods + prep	Traditional lab + optional AI	Manual or digital logging

EXPERIENCE BEFORE EQUATION • MEASURE FIRST • QUESTION THE PREMISE

TEACHER AT A GLANCE

What this lesson is really about

The physics matters. So does noticing when a seemingly reasonable comparison is not actually equivalent.

Instructional purpose

Students compare the cooling behavior of water, sodium-chloride brine, and calcium-chloride brine. The lesson begins as a familiar cooling-curve investigation, then exposes a deeper experimental-design issue: adding solute changes both composition and total mass. The optional AI extension asks whether an intelligent reasoner notices that problem before being shown the data.

Physics target

Specific heat, total thermal capacitance, heat transfer, Newton-style cooling, dissolution energy.

Scientific-practice target

Control variables, distinguish a property from a system quantity, identify hidden variables, redesign a comparison.

AI-literacy target

Evaluate reasoning, not eloquence. Ask whether the agent checks assumptions and distinguishes data from premise.

Engineering/SDL target

Use simple sensors or automated logging without letting instrumentation replace experimental judgment.

Learning outcomes

- Explain the difference between specific heat capacity, c, and the thermal capacitance of an entire sample, approximately m x c.
- Collect and graph temperature-versus-time data under controlled conditions.
- Identify why “250 g water” and “250 g water + 90 g salt” are not equal-mass systems.
- Distinguish the heat released or absorbed during dissolution from the later cooling behavior of an already prepared solution.
- Propose a redesigned experiment that more cleanly isolates composition, mass, or another chosen variable.
- Optional: assess whether an AI reasoner detects a flawed premise without being explicitly told where to look.

Teacher stance

Do not reveal the trap too early. Let prediction, measurement, and discrepancy create the need for better reasoning.

SCIENCE BACKGROUND

The physics beneath the comparison

Two quantities students often collapse into one

Specific heat capacity: c

Energy required per unit mass per degree of temperature change.

Sample thermal capacitance: $C \approx m \times c$

Energy required to change the temperature of the whole sample by one degree.

A solution can have a lower specific heat than water yet still have a comparable - or even larger - total thermal capacitance if there is more of it. In the canonical challenge condition, the salt samples contain 250 g of water plus 90 g of solute, so their total mass is about 340 g, not 250 g.

A useful first-order model

Heat loss: $\dot{Q} \approx hA(T - T_{amb})$

The instantaneous heat-loss rate depends on the temperature difference and heat-transfer conditions.

Time scale: $\tau \approx C / (hA)$

Cooling rate depends on total thermal capacitance and on how effectively heat leaves the sample.

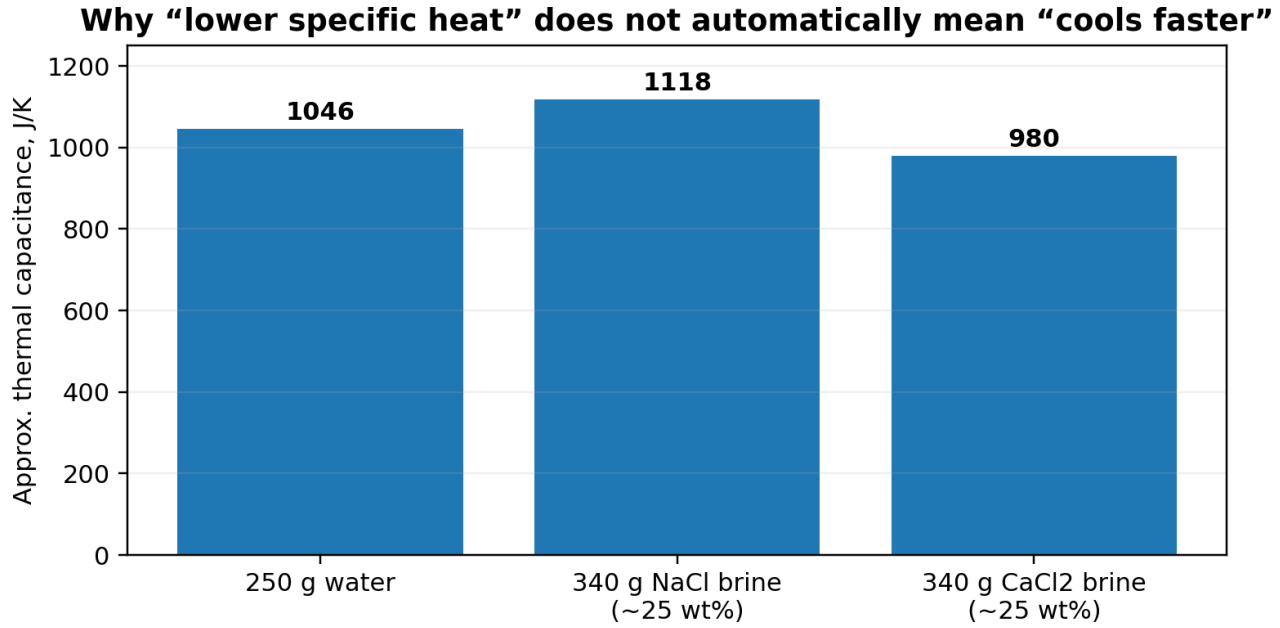
This model is deliberately simple. Real classroom samples can differ in evaporation, viscosity, natural convection, surface temperature, container coupling, probe placement, and heat transfer to the air. Those effects are not “noise to ignore”; they are part of the reason a clean experiment needs careful controls.

Why NaCl and $CaCl_2$ are pedagogically useful

- NaCl brine has a lower specific heat than pure water, but adding substantial salt also increases total sample mass.
- $CaCl_2$ brine has a still lower specific heat at comparable mass fraction, making it a useful contrast.
- Dissolving $CaCl_2$ in water is strongly exothermic. That is a separate thermal event and must not be confused with the later cooling curve.
- Commercial “rock salt” is mostly NaCl and should behave broadly like crude NaCl; commercial calcium-chloride ice melt can provide an accessible $CaCl_2$ source if its label and purity are documented.

THE REASONING TRAP

Lower c does not automatically imply faster cooling



Illustrative only: uses ASHRAE 15 °C values at 25 wt%; heat transfer coefficients and evaporation also matter.

The bars above are an explanatory approximation, not expected experimental results. They use published 15 °C brine specific-heat values near 25 wt% to illustrate the accounting problem. A 340 g NaCl brine can have a total thermal capacitance comparable to or larger than 250 g of pure water even though its specific heat per kilogram is lower. CaCl2 at similar concentration can move the other direction.

Canonical trap

A student - or AI - may reason: “salt water has lower specific heat, therefore it must cool faster.” That inference skips the changed mass and treats c as if it were the heat capacity of the whole vessel.

The deeper scientific question

The lesson is not designed to prove that one liquid always cools faster than another. It is designed to ask what must be controlled, what is actually being compared, and whether a prediction follows from the stated evidence. The experimental result is allowed to surprise the class.

A note on the 90 g NaCl condition

Ninety grams of NaCl added to 250 g of water produces a nominal 26.5 wt% mixture if all salt is in solution. That is near the room-temperature saturation region, so complete dissolution should be checked. If crystals remain, record that fact and treat the solid as part of the physical system rather than pretending it is a homogeneous solution. For a cleaner solution-only variant, use a lower NaCl mass.

Accessible materials, professional controls

Core materials - per group or demonstration station

Qty	Material	Teacher note
3	Identical heat-resistant vessels, 350-500 mL	Same geometry and surface area
3	Temperature probes or thermometers	Matched probes preferred
1	Digital balance, ±1 g or better	Mass-based setup
750+ g	Water	Same source for all conditions
90 g	Sodium chloride	Lab salt, table salt, or documented rock salt
90 g	Calcium chloride product	Pellets/flakes labeled CaCl2; ideally 90%+
1	Water bath or controlled warming method	Bring all samples to same start temperature
1	Timer / logger / SDL data path	15-30 s logging ideal; manual 1-2 min acceptable

Calcium chloride safety

PPE and preparation

Wear splash goggles and gloves. Calcium chloride is an eye irritant, and dissolution in water is exothermic. Teacher preparation is preferred for younger students.

- Use cool or room-temperature water for preparation. Add CaCl2 gradually to water while stirring; do not dump a large charge into hot water.
- Use an open, heat-resistant container during mixing. Do not seal a warm dissolving mixture.
- Keep dry pellets and dust away from eyes; wash hands after handling. Follow the product SDS and the school chemical-hygiene plan.
- After the solution has been prepared, allow it to return to a controlled starting condition before beginning the cooling comparison.
- Chloride brines can be corrosive to some metals. Rinse probes and equipment promptly after use.
- Follow local/school rules for disposal. Do not improvise disposal instructions when using a commercial product with additives.

Consumer-source rule

If using ice melt, select a product whose ingredient statement identifies calcium chloride. Avoid generic “blend” products unless the mixture itself is intentionally part of the investigation. Record brand, stated purity, form, and lot if available.

TEACHER PREPARATION

Separate solution-making from cooling behavior

Prepare the three challenge samples

Condition	Composition	Approx. total mass	Purpose
A - Water	250 g water	250 g	Baseline
B - NaCl challenge	250 g water + 90 g NaCl	~340 g	Intentional mass change; near saturation
C - CaCl2 challenge	250 g water + 90 g CaCl2 product	~340 g	Strong heat of dissolution; record purity

Preparation sequence

1. Label vessels and record empty-vessel masses if your class will perform a full energy accounting.
2. Prepare the NaCl solution. Stir until dissolved as completely as practical; record any persistent crystals.
3. Prepare the CaCl2 solution separately. Add the solid gradually to water while stirring. Observe the temperature rise as a distinct phenomenon, not as cooling-curve data.
4. Allow all samples to approach the same general temperature before controlled warming.
5. Use a water bath to bring all three samples to the same chosen starting temperature, typically about 50-60 °C. Do not rely on “same heating time.”
6. Place vessels in the same environment, with matched probe depth and position. Use identical loose covers if evaporation control is desired. Do not seal heated vessels.
7. Start the cooling run only when all samples are within the teacher-selected start tolerance, for example ±0.5 °C.

> **Important**
>
> The dramatic temperature rise when CaCl2 dissolves is worth observing - but it answers a different question. Mixing enthalpy and cooling behavior in the same run confounds the experiment.

Suggested logging

For digital probes, record every 15-30 seconds for 15-20 minutes. For manual thermometers, 1-minute or 2-minute intervals are sufficient. Record ambient temperature at the beginning and end.

INVESTIGATION A

The intentionally naive comparison

Students encounter the simple claim before the experimental design is repaired.

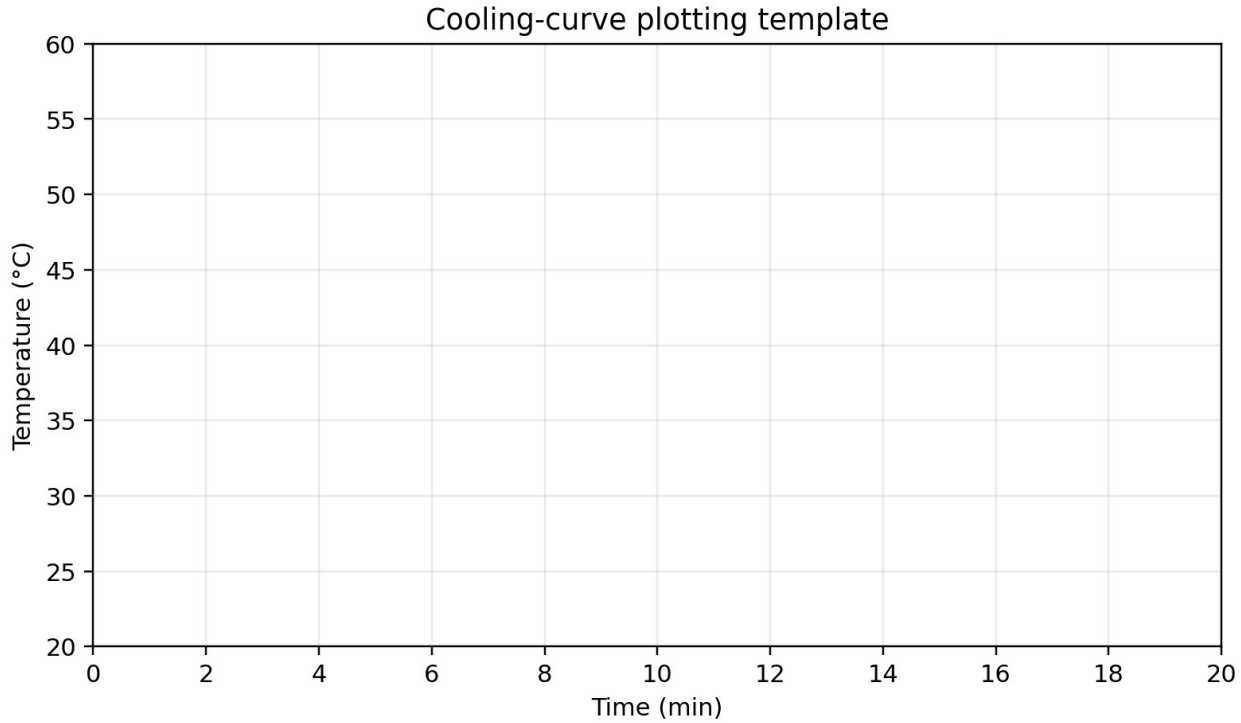
Before data collection: require a prediction

Ask each student or team to rank the three samples from fastest to slowest expected temperature decrease. Require one sentence of reasoning and one variable they believe must be controlled. Do not yet teach the mass trap.

> **Prompt**
>
> “Salt solutions have lower specific heat than pure water. What do you predict will happen when these three samples cool from the same starting temperature? Explain your reasoning.”

Run the experiment

1. Confirm identical starting temperature within the chosen tolerance.
2. Start timer/logging simultaneously for all conditions.
3. Do not stir during the cooling run unless stirring is explicitly standardized for every vessel.
4. Continue until the samples have cooled substantially toward room temperature, typically 15-20 minutes.
5. Graph temperature versus time for all three conditions on the same axes.
6. Ask students to describe what the graph shows before asking them to explain why.



Do not over-promise the result

The exact curve order is not guaranteed. Cooling depends on more than specific heat. The educational success criterion is whether students use the data to revisit the model and identify the variables that actually determine the observed cooling time.

INVESTIGATION B

Repair the comparison

Turn the hidden-variable discovery into experimental design.

The reveal

Ask first

“How much material is actually in each vessel?” Then write 250 g, 340 g, 340 g on the board. Ask whether the first experiment isolated composition.

At this point students should distinguish the intensive property c from the extensive system quantity C. The first experiment is not “wrong”; it is answering a different question than many people initially think it is answering.

Three legitimate redesign choices

Equal total mass

Prepare the same total mass in each vessel. Example: 250 g total sample, with brines made to a chosen product mass fraction.

Equal water mass

Keep 250 g water in every vessel and deliberately study what happens when dissolved material is added. This is the original challenge design.

Equal concentration

Hold brine mass fraction constant while comparing NaCl with CaCl2. This isolates salt identity more cleanly, but their molar concentrations differ.

Equal moles of solute

Useful for a chemistry-focused extension. It is not automatically the fairest thermal comparison; state what variable the redesign is intended to isolate.

Recommended second run: equal total sample mass

A practical classroom redesign is 250 g total in every vessel. For a nominal 25 wt% brine using pure material, that would be 187.5 g water + 62.5 g solute. With commercial CaCl2, use the product label to decide whether you are comparing equal product mass fraction or correcting for stated CaCl2 purity. Either choice is acceptable if it is documented.

Experimental-design principle

There is no single magical “fair” comparison. A fair test is one whose controlled variables match the specific causal question being asked.

ANALYSIS & TEACHER KEY

What strong reasoning should contain

Evidence-to-explanation sequence

Step	Student task	Teacher look-for
1. Describe	Which curve falls fastest at each stage? Do curves cross? What is the initial slope?	No causal language yet.
2. Quantify	Compare ΔT over a fixed interval, slope, or fitted time constant.	Use the same metric for all samples.
3. Account	Compare total mass and approximate m x c, not c alone.	This is the central trap.
4. Challenge	Ask whether hA, evaporation, convection, probe position, or undissolved solid changed.	Cooling is a system process.
5. Redesign	State which variable the next experiment will isolate.	A better experiment follows from the discrepancy.

Approximate property anchors

Published brine data provide useful scale, not an answer key for the exact classroom run. At 15 °C, ASHRAE lists specific heat near 3.29 kJ/(kg·K) for 25 wt% NaCl brine and about 2.88 kJ/(kg·K) for 25 wt% CaCl2 brine, compared with about 4.18 kJ/(kg·K) for water. When sample mass is also included, the ordering of whole-sample thermal capacitance is less obvious than the ordering of specific heat.

Expected misconceptions

“Lower c means faster cooling.”

Incomplete. Cooling time depends on whole-sample thermal capacitance and heat-transfer conditions.

“Same water means same sample.”

No. Adding 90 g of salt changes total mass and composition.

“The CaCl2 got hot, so it retains heat.”

The initial temperature rise is dissolution enthalpy; it must be separated from the later cooling comparison.

“One graph proves the material property.”

No. A cooling curve reflects both material properties and the experimental heat-transfer pathway.

Best teacher answer

The experiment is most valuable when the data force students to ask a better question. The target is not a predetermined curve order; it is disciplined causal reasoning.

OPTIONAL AI EXTENSION

Can the reasoner find the flaw?

AI is a participant in the reasoning test - not the subject of the physics lesson and not merely an automation tool.

Protocol: keep the reveal staged

Stage	Teacher action	Reasoning behavior to score
A. Premise test	Ask for a prediction before sharing data.	Does the AI notice that adding solute changes total mass?
B. Mass disclosure	State the exact masses: 250 g water; 250 g water + 90 g salt.	Does it revise its argument and invoke m x c?
C. Data confrontation	Provide the measured curves without explaining them.	Does it distinguish observation from explanation?
D. CaCl2 twist	Mention that the CaCl2 solution became hot during preparation.	Does it separate heat of dissolution from cooling behavior?
E. Redesign	Ask for a fairer next experiment.	Does it say what variable is being isolated and why?

Suggested AI prompts

- “I have 250 g of water and another sample made from 250 g of water plus 90 g NaCl. Both begin at the same temperature. Which should cool faster, and why?”
- “What assumptions in your answer could make that prediction fail?”
- “Here are the temperature-time data. Explain what changed in your reasoning.”
- “Now replace NaCl with calcium chloride ice melt. What new physical effect must be separated from the cooling run?”
- “Redesign the experiment to isolate composition rather than accidentally changing multiple variables.”

> **Important distinction**
>
> A fluent answer is not evidence of sound reasoning. Score whether the agent checks the premise, accounts for system mass, handles uncertainty, and proposes a falsifiable redesign.

What not to do

- Do not tell the AI in advance that there is a trick.
- Do not grade it on whether it predicts the final curve order correctly without experimental parameters.
- Do not let AI-generated explanation replace student measurement or graph interpretation.
- Do not present this as “AI versus student.” The productive question is what kind of reasoning behavior appears, under what information conditions.

ASSESSMENT

A rubric for physics, experimental design, and AI reasoning

Dimension	1 - Emerging	2 - Developing	3 - Proficient	4 - Advanced
Specific heat vs heat capacity	Treats c as the whole story	Recognizes mass matters	Uses C≈m c correctly and explains implication	Uses C plus heat-transfer pathway to qualify prediction
Experimental control	Lists few controls	Identifies start T / container	Identifies mass, geometry, probe, ambient, evaporation	Links each control to the causal claim
Dissolution vs cooling	Conflates events	Notices CaCl2 warms	Separates solution preparation from cooling run	Uses the distinction to redesign procedure
Use of evidence	States expected answer	Describes graph	Quantifies curve behavior	Uses discrepancy to revise model
AI evaluation - optional	Accepts fluent answer	Checks one claim	Tests assumptions and asks for uncertainty	Treats AI reasoning as a claim that can itself be experimentally probed

Exit ticket

1. In one sentence, explain why lower specific heat does not by itself prove faster cooling.
2. Name one hidden variable in the original 250 g water versus 250 g water + 90 g salt comparison.
3. Why must a CaCl2 solution be allowed to reach the chosen starting temperature before its cooling curve begins?
4. Design one improved comparison and state exactly what variable it isolates.
5. Optional AI: What evidence would convince you that an AI reasoner had actually checked the experimental premise rather than merely produced a plausible explanation?

Extension for advanced classes

Fit a first-order cooling model T(t) = Tamb + (T0 - Tamb)e^(-t/τ) and compare fitted τ values. Then discuss why τ is a property of the sample-plus-container-plus-environment system, not a pure material constant. Advanced students can estimate C = m c and infer an effective hA from τ ≈ C/(hA), while explicitly stating the model assumptions.

SDL IMPLEMENTATION

Instrumentation should expose reasoning, not hide it

Minimal implementation ladder

Level 0 - Manual

Three thermometers, timer, notebook. Fully valid science.

Level 1 - Digital probes

Three temperature sensors logging to a laptop or phone.

Level 2 - SDL logger

A microcontroller or remote sensor reports human-readable temperature/time data to a dashboard.

Level 3 - AI-connected SDL

The same live data can be made available to an AI reasoner - while the teacher controls what context and assumptions are disclosed.

Recommended data record

Time (min)	Water °C	NaCl °C	CaCl2 °C	Notes
0				
2				
4				
6				
8				
10				
12				
14				
16				
18				
20				

Metadata worth saving

- Container material, dimensions, and cover condition
- Exact water and solute masses; product purity/ingredient label
- Start temperature tolerance and ambient temperature
- Probe IDs, calibration check, and immersion depth
- Whether samples were stirred during cooling
- Visible undissolved NaCl, evaporation, spills, or other departures from protocol

SDL principle

Automation earns its place when it makes the experiment more observable, repeatable, and inspectable. It should never conceal the variables that make the scientific reasoning meaningful.

REPRODUCIBLE STUDENT CHALLENGE

One-page prompt for prediction, observation, and redesign

The challenge

Three samples begin at the same temperature in identical containers:

- A: 250 g water
- B: 250 g water + 90 g sodium chloride
- C: 250 g water + 90 g calcium-chloride ice-melt product

Before the experiment

Rank A, B, and C from fastest to slowest expected cooling.

Give your physical reason.

List three variables that must be controlled.

What evidence would make you change your prediction?

After the experiment

What did the graph actually show?

What was the most important hidden variable in the original comparison?

How would you redesign the experiment to isolate the variable you care about?

Optional AI comparison

Ask an AI the same first prediction question before giving it data or hints. Save its original answer. After the experiment, compare how your reasoning and the AI reasoning changed.

Property data and safety sources

Technical references

ASHRAE Handbook - Fundamentals, Chapter 31: Physical Properties of Secondary Coolants (Brines).

Tables for NaCl and CaCl2 brines provide specific heat, density, freezing behavior, and commercial-grade notes. The guide uses these values only as approximate property anchors; classroom cooling curves depend on the full heat-transfer system.

MilliporeSigma / Sigma-Aldrich Safety Data Sheets for calcium chloride and calcium chloride dihydrate.

Classifies calcium chloride products as eye irritants and documents PPE/handling considerations. Always follow the SDS for the exact classroom product.

Commercial calcium-chloride ice-melt product label / SDS.

Record stated calcium-chloride percentage and additives. Commercial products are appropriate for an accessibility-focused classroom investigation when their composition is documented; they are not reagent-grade standards.

Source access: ASHRAE Handbook online, Chapter 31; Sigma-Aldrich SDS portal.

Teacher note on claims

Do not teach a fixed ordering such as “CaCl2 always cools fastest.” The experiment is intentionally sensitive to system mass and heat-transfer conditions. The robust conclusion is methodological: specific heat is one part of the model; the whole sample and the whole heat-transfer pathway determine the observed cooling curve.

Teacher note on AI

The optional AI activity is strongest when the AI is treated as another reasoning participant whose assumptions can be tested. The distinctive question is not whether an AI can calculate a property value, but whether it detects that the experimental premise may be underspecified or misleading.

> **Enduring idea**
>
> Good experimental science does not merely calculate from a premise. It checks whether the premise deserves to be calculated from.